Trade and innovation: evidence from Latin America

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Abstract

The rise of China in international trade has induced significant trade exposure to many economies worldwide. This paper explores the China shock as an exogenous source of increased import competition, input access and market expansion in Latin America. Drawing on cross-country patent data and trade data decomposed through additional input-output statistics, this study sheds light on an additional form - input access - of trade exposure in the context of trade and firm innovation in emerging economies. The current literature offers mixed evidence on the impact of the China shock, and there is limited evidence on the effect of an increase in imported intermediates. This study finds that input access was the strongest channel of the China shock on firm innovation, inducing particularly firms in high-technology sectors to increase their patenting activity.

Keywords: Latin America, China shock, innovation, input-output

JEL classification: F14, F6, L60, O31, O34

1 Introduction

The rapid integration of China into the global economy represents one of the most consequential developments in trade in the last quarter century. This episode, reflecting the emergence of China as a dominant player in global manufacturing after its accession to the World Trade Organization in late 2001, has been often referred to as the "China shock" in the economics literature (Autor et al., 2016). As such, it was perceived in all channels of trade exposure: increasing import competition, access to imported inputs, and market access to exports. In just six years, from 2002 to 2008, China's weight as an exporter nearly doubled, representing from 4.1% to 7.6% of world exports. As an importer, its weight rose from 3.7% to 5.9% in the same period.[1]

Analyses of the repercussions of the China shock on firms have focused on developed countries (Aghion et al., 2024; Ahn et al., 2018; Autor et al., 2020; Bloom et al., 2016). Evidence on emerging economies is significantly more limited, particularly regarding the impact of the China shock on firm performance. Most studies focus on firm productivity and leverage trade liberalization periods other than the rise of China in world trade (Amiti and Konings, 2007; Bas and Ledezma, 2010; Bustos, 2011; Schor, 2004).

This study aims to add to the smaller, but growing body of evidence on the experience of emerging countries during the China shock. It has the aim of understanding how China's rise impacted firm innovation in Latin America. Innovation, although a key microchannel of the relationship between trade and growth (Melitz and Redding, 2021), is a type of firm outcome that is less common in the literature, compared to productivity. This might be partially explained by the fact that patenting activity, a common measurement of innovation, is much less frequent in emerging economies. To improve on this limitation, I perform a cross-country study that focuses on trade exposure on country-industry pairs for Argentina, Brazil and Mexico, which collectively account for over 85% of all patenting activity in Latin America between 1996 and 2008. This article also aims to fill a literature gap by providing evidence on the China shock's impact on emerging economies across all three forms of trade exposure, meaning it disentangles import penetration from input access using difference sources of input-output data. The final dataset reflects firm patent and citation counts combined with bilateral country-industry (4-digit) trade estimates.

The literature on the impact of input access from the China shock on firm innovation is par-

[1] Figures calculated with World Bank data.

ticularly scarce. The closest study to this paper is Aghion et al. (2024), which neatly disentangles import competition and input access using French firm-level data from accounting and customs statistics. They find opposite effects, where import competition was detrimental to firm innovation, and the input supply shock had a positive (although not always significant) impact on patenting. Analyzing the impact of the China shock on manufacturing employment in the United States, Acemoglu et al. (2016) make use of a national input-output table for 1992 in a similar strategy to this study to capture spillover effects across supply chains, finding that downstream shocks (to buyers of a firm's inputs) have particularly negative effects on employment.

In this study, I find that the strongest effect of the China shock on Latin American patenting was the increased access to Chinese inputs, which had robust and significant innovation-inducing effects. This is particularly true for firms in high-technology sectors, or those who source their inputs from high-technology sectors. I also find that import competition had a weaker, but robust negative effect on firm patenting activity. Similar coefficients were identified for the market access channel, although it is substantially less impactful in the context of Latin America, where the China shock mostly created demand for commodities, rather than manufactured goods.

The remaining part of this paper is organized in the following way. Section 2 reviews some theoretical concepts and the current empirical literature on trade liberalization and innovation. Section 4 outlines data sources and estimations, as well as descriptive statistics of the final dataset. Section 5 describes the empirical strategy, and section 6 presents results. Section 7 concludes.

2 Theoretical motivation

Innovation can take multiple forms for firms. While its main input is generally agreed to be investment in R&D, its outcome, if successful, can be the adoption of new technology processes, upgraded product quality, or a new set of products altogether. The intellectual property of these outcomes is often protected by patents.

In a broader context, innovation is inserted in the literature of firm performance. Most studies analyzing the linkages between trade and firm performance focus on productivity (Shu and Steinwender, 2019). In this regard, innovation will ultimately affect productivity, but productivity gains can arise from processes that do not involve innovation. Additionally, the translation of innovation into added productivity is slow, which is why measuring innovation directly through outcomes such

as patent counts has been a common approach in the literature (Akcigit and Melitz, 2022).

Investment in R&D, which may or may not result in a successful innovation, is determined by firms as a function of expected future profits from this potential innovation. This investment often requires high levels of fixed costs and therefore benefits from economies of scale. Aghion et al. (2024) document multiple innovation premia, showing that French firms that export and innovate are substantially bigger and more productive than non-innovating exporters. In turn, these future profits are dependent on product market competition, which itself can be significantly affected by trade liberalization (Melitz and Redding, 2021).

Different types of trade exposure can have different effects on firm performance. Trade liberalization can mean both an increased market access for a given firm (increased sales through exports) and an inflow of imports in its domestic markets. A key point to consider is that a firm can act as a seller of products that can be either final goods and services or intermediates, or as a buyer of intermediate products from upstream in its supply chain. Consequently, for a given firm, an inflow of imports can imply two different outcomes: where imports are of similar products to those the firm sells, it faces import competition; where imports are of inputs the firm buys, it enjoys increased input access (Shu and Steinwender, 2019). The next three subsections briefly discuss the literature on these three distinct forms of trade exposure and their effects on firm performance, and particularly innovation.

2.1 Market access effect

Increased export opportunities primarily lead to increased firm revenue as its market expands to consumers abroad: including both other firms (exports of intermediates) or households or the government, for example (exports of final products). This can help spread fixed costs of R&D investment and the expected value of successful innovations (Geng and Kali, 2021).

There is extensive evidence of a positive exports shock on firm performance and innovation. Ahn et al. (2018) find a positive effect of rising exports to China on Korean firms, and Coelli et al. (2022) argue more broadly that the decline in tariffs through the 1990s was an important driver of global innovation in the period. Focusing on emerging economies, and in particular Latin America, Bas and Ledezma (2010) show that a reduction of export barriers is positively correlated to plant productivity in traded sectors in Chile. Building on the seminal model of trade with heterogeneous

firms from Melitz (2003), Bustos (2011) endogenizes the decision of the firm to upgrade technology, predicting that falling trade barriers can induce exporting firms to upgrade technology. She finds that, by lowering the cutoffs of firm entry in the export market and to upgrade technology, as well as by increasing firm revenue, lower export barriers have a positive effect on firm spending on technology.

However, there is a nuance to the market access mechanism that is typically synonym to export liberalization. Increased market access refers to the intensive margin of incumbent exporters, but does not reveal the effects on the extensive margin that is predicted by the theory of trade with heterogeneous firms. Analyzing French manufacturing firms, Aghion et al. (2024) find that exogenous growth shocks in export destinations lead to increased patenting. Since this impact is entirely driven by firms in the top half of labor productivity, they formalize this heterogeneity in a model that also accounts for an induced competition - incentive for entrants in the domestic market, building on the model by Melitz and Ottaviano (2008) - effect from export access, which is detrimental to innovation. This effect can dominate and lead to negative innovation for initially least productive firms. Since this affects all firms in the domestic market, non-exporting firms can also face particularly negative effects with respect to performance (Baldwin and Gu, 2009). In the case of Brazil, Ghizzi (2021) notably finds an overall negative effect of export liberalization on firm innovation.

2.2 Import competition effect

Beyond the context of international trade, the nexus between competition and innovation has been part of a long-standing discussion in economics. On the one hand, competition can be detrimental to innovation by simply restricting the size of firms, which weakens their economies of scale (Schumpeter, 1943). This Schumpeterian effect is contrasted with the escape-competition effect (Arrow, 1962), where more competition induces incumbents to increase R&D due to a decrease of their pre-innovation rents, thereby raising incremental returns from innovation. Aghion et al. (2005) synthesize these competing ideas and point to a nonlinear relationship between competition and innovation, depending on the pre-shock productivity dispersion of industries: in neck-and-neck competition, the escape-competition effect would dominate, while in leader-and-laggard competition (laggards far from the productivity frontier), the Schumpeterian effect would prevail in a shock of additional competition, resulting in less firm innovation.

While competition plays a role in market access, its key mechanism in the context of trade is import competition. As expected, much like theory suggests, empirical findings for this form of trade exposure have yielded mixed results. A first grup of studies find a positive relationship between import competition and innovation in Europe (Bloom et al., 2016), Korea (Ahn et al., 2018), and Japan (Yamashita and Yamauchi, 2020), highlighting the escape-competition effect. Conversely, studies for the United States (Autor et al., 2020), Canada (particularly on process, rather than product innovation) (Yang et al., 2021), France (Aghion et al., 2024), and Brazil (Ghizzi, 2021; Moreira et al., 2023). Regardless of the direction of the effect, most of these studies provide widespread evidence that larger, more productive firms are relatively better off, or at least less worse off, than laggard firms when facing an import competition shock.

2.3 Input access effect

As outlined, a firm may play opposing roles in a given market, namely as a seller of either final products to households or governments or of inputs to other firms, otherwise as a buyer of inputs from other firms. Current theory suggests that, unlike import competition (a horizontal shock of similar goods), the access to more imported inputs (a vertical shock of goods used in the focal firm's production) as a consequence of trade liberalization should have positive consequences on firm performance and innovation. Access to intermediate goods can lead to increased firm efficiency through knowledge spillovers (Grossman and Helpman, 1991) and increased profits through cheaper inputs (Shu and Steinwender, 2019), particularly from emerging economies like China in the early 2000s.

The literature on input access has grown more recently due to increased access to input-output and firm-level data required to disentangle inputs from overall imports. The empirical evidence on firm productivity is clear, with multiple case studies showing the positive effects of input access on emerging economies (Amiti and Konings, 2007; Schor, 2004; Topalova and Khandelwal, 2011). There is also evidence on its positive, albeit often less significant, impact on innovation, measuring patenting (Aghion et al., 2024; Bloom et al., 2016), product innovation (Bas and Paunov, 2021; Colantone and Crinò, 2014), and technology adoption (Bas and Berthou, 2016).

3 An overview of the China shock in Latin America

Before discussing descriptive statistics, it is worth taking stock of how the China shock unfolded in Latin America. Namely, the context of Latin American economies in the late 1990s and early 2000s. The region had gone through a substantial process of trade liberalization through the 1990s, substantially reducing their average tariffs.[2] This had significant consequences for the industrialization policies that had been led since at least the 1940s, focusing on import substitution of protection of nascent industries. A process of deindustrialization in Latin America had therefore already begun at the time of the China shock, where the trend was inverse of emerging economies such as those of East Asia (Duran et al., 2008).

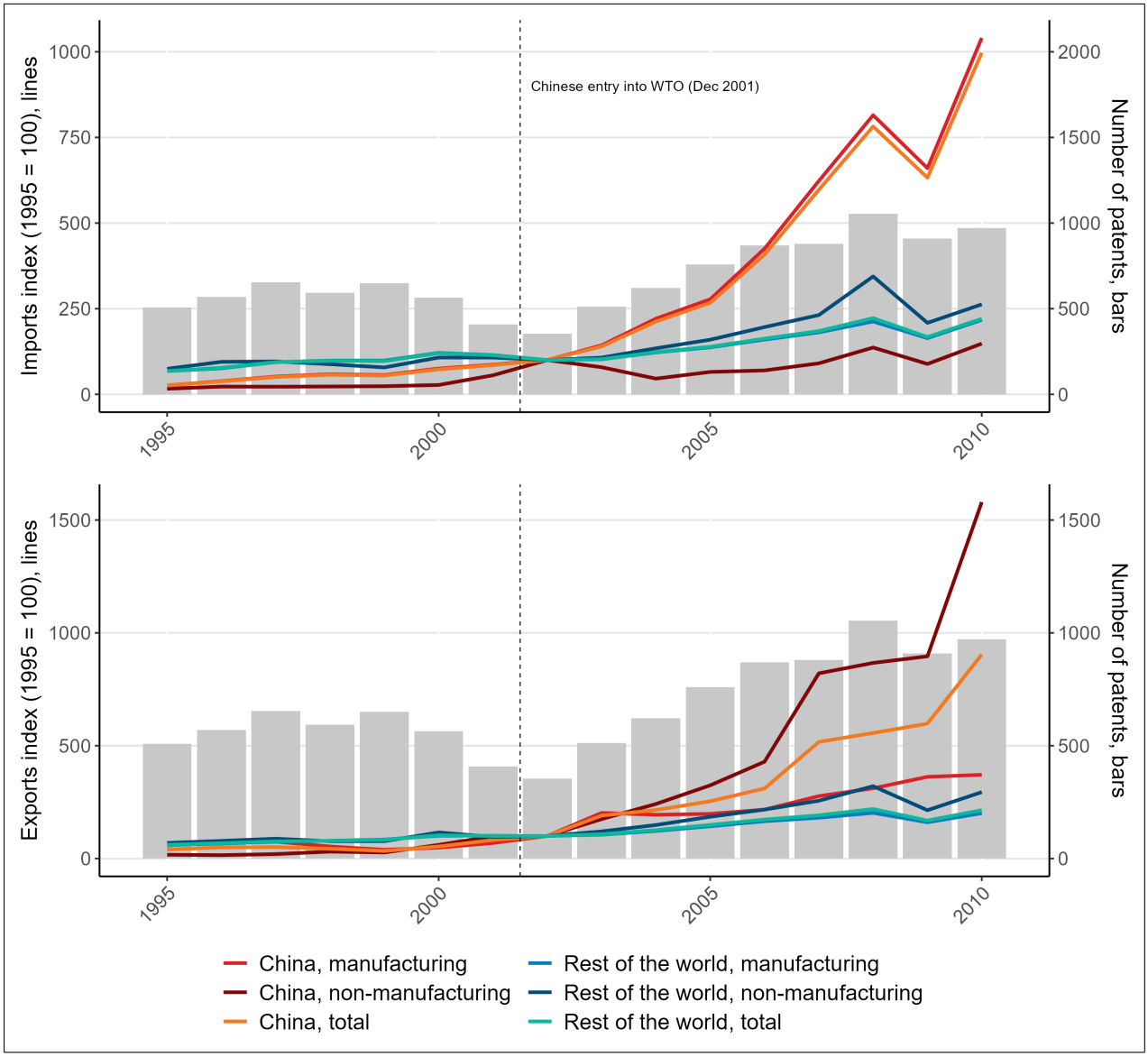
Additionally, the late 1990s and early 2000s were marked by a series of currency crises in emerging economies, including Mexico, Brazil and particularly Argentina. The Argentinian convertibility crisis involved great political, social and economic unrest, ultimately leading to a real GDP of -13% in between the last quarter of 1998 and the last quarter of 2001, although recovery was swift (Salvia, 2015). These instabilities had an effect on Latin American innovation before the China shock, which was trending downwards. Proportionally, Argentina had the highest decrease in this period, but Brazil also experienced a significant drop in patenting activity between 2000 and 2002.

Figure 1 presents the evolution of patenting in Latin America (Argentina, Brazil, and Mexico) between 1995 and 2010, together with indexes portraying the growth of imports and exports in the region. The latter are split by partner (China and the rest of the world, ROW) and sector (manufacturing and non-manufacturing). There is a temporary downturn in 2009 due to the financial crisis, where trade in China recovered faster than trade with ROW. To avoid capturing this episode, I define the period of the China shock from 2002 until 2008. The discussion below will focus on the growth between 2002 and 2008.

A first message from this figure is how China deeply outpaced ROW as an importer from and exporter to Latin America. However, this evolution is heterogeneous when considering different flows and sectors. In the case of manufactured imports, it outgrew ROW by a factor of nearly seven (715% compared to 113% growth, respectively). Conversely, for non-manufacturing imports, it was outgrown by the rest of the world, suggesting the influx of goods from China during the shock was overwhelmingly that of manufactured goods.

[2]The average (unweighted) applied tariff in Latin America fell from 40% in 1985 to around 13% in 1995 (Duran et al., 2008)

Figure 1: The rise of China and innovation in Latin America, 1995-2010



Source: Author's calculations based on BACI (CEPII), combined with BTDIxE (OECD).
Note: Manufacturing corresponds to ISIC Rev.4 industries 10 to 33. Non-manufacturing corresponds to the residual trade, mainly from the agriculture and extractive sectors.

A crucial narrative takes shape when observing the evolution of exports to China: the China shock for Latin American exports translated mostly into demand for non-manufactured goods. This stands in contrast with most studies analyzing the shock for developed countries, which did also see a substantial increase in Chinese demand for their manufactured goods. Although the growth of Latin American manufactured goods exports outpaced that of ROW by a factor of two, it did so by a factor of nearly four for non-manufactured goods. The main consequence of this is that, while the rise of China's weight as a trade partner in Latin American manufactured imports is clear and

pronounced, it is not so for the region's manufactured exports. There is evidence that demand side of this commodities-for-manufactures trade boom had positive welfare effects in Latin America, such as wage growth for workers in agriculture (Costa et al., 2016). However, in the context of this study's outcome, firm patenting - essentially all of which is manufacturing -, this translates into less explanatory power of the China shock regarding the increased market access mechanism in Latin America.

4 Data

With the goal of analyzing the impact of the China shock on Latin America across the three described forms of trade exposure, I combine firm-level patenting data with trade data at the 4-digit industry level. The trade data corresponds to a novel dataset of bilateral trade data that also accounts for the trade of intermediates between China, Latin America and the rest of the world. To achieve this, I combine product-level trade data with a well-established world input-output table dataset, benefitting from extended industry granularity based on national supply-use tables. The final data runs from 1996 through 2008, accounting for the pre-shock period of 1996-2001, and the shock period from 2002 to 2008.

As it aims to draw insights from patenting activity throughout Latin America, the database is cross-country and covers the three main patenting countries in the period: Brazil, Argentina and Mexico. According to PATSTAT data, these countries together represent approximately 85% of all granted patents and innovating firms in Latin America in the period. This approach enables further analysis in a region where patenting is much sparser than in developed economies. Doing so maximizes the number of observed markets (in this study, country-industry pairs) distinctly affected by the China shock.

4.1 Patent data

I retrieve patent data from the European Patent Office's Worldwide Patent Statistical Database (PATSTAT), which offers bibliographic data, family links and technology classification of more than 100 million patent documents worldwide since the mid-1960s. It enables the construction of the sample required for this study, including information on the applying firm: its country and date of application; and the application: if it was granted, its technology area (IPC code), its number

of associated citations, and its INPADOC family. The latter is important to group applications of similar technical content to determine unique patents that might have entailed multiple applications in different countries. In this paper, I consider an innovation to be an INPADOC patent family that includes at least one granted application.

A well-known limitation of patent data is that the firm names entered on patent applications frequently contain different abbreviations or misspellings, which means a single innovating firm might be misinterpreted as different firms innovating in different sectors. To have robust data on patenting at the firm level, I thus engage a name matching methodology which employs regular expression algorithms and a thorough analysis of company names. While the former minimises the need for manual analysis, the latter is required as countries might have different versions a business acronym (i.e., Ltd.), and a substantial number of misspellings are unpredictable. I remove these basic acronyms, city and country names, punctuation, and whitespaces to then analyse similar names manually. I perform a final match on other misspelled names with the help of a string distance metric, namely the Jaro-Winkler distance, to support the identification of nearly identical firm names.

To infer the patenting firm's industry, I borrow the methodology from Coelli et al. (2022) and use the information on the technology classification of its patents. Aghion et al. (2016) show that this patent-based scheme can be a legitimate measure since it reflects the firm's expectations of where their market will be, and that firms' patent weights are highly correlated with their sales weights. Therefore, unlike the counting of innovative patents, for weighing purposes I can extend the counting to also non-granted applications, as well as include a pre-period to my baseline analysis, with a span from 1980 to 2007. As PATSTAT already converts IPC codes into ISIC 4 industries, I define:

$$\theta_{f,i} = \frac{x_{f,i}}{\sum_k x_{f,k}} \tag{1}$$

where $x_{f,i}$ is the number of applications issued by firm f in industry i during the period. I then establish a firm's industry by identifying the maximum weight across all its applications.

One of the concerns of using patent production is that not all new patents are a result of innovation, since higher competition can result in more protection of already existing intellectual property. However, Coelli et al. (2022) have recently provided evidence that this "lawyer effect" was not predominant in the increased patent production of the late 1990's trade liberalisation by demonstrating that patent quality (citations) actually increased in the same period because of more

market access. I provide additional evidence on the absence of the "lawyer effect" when analyzing the impact the China shock on Latin American innovation.

4.2 Trade data

To construct a measure of exposure to the Chinese rise as an exporter of final and intermediate goods, I face the challenge of building a disaggregated (ISIC Rev.4 4-digit) bilateral trade dataset that provides not only information on the exporting industry, but also the importing industry. Since cross-country trade databases usually provide only information for the country, but not industry destination, extra sources are required to estimate this missing dimension.

My main source is the BACI database developed by CEPII (Centre d'Etudes Prospectives et d'Informations Internationales). An advantage of this database is that it harmonizes UN COMTRADE data by matching the value between CIF (cost, insurance, and freight) imports and FOB (free on board) exports. This database is at the product level, which is aggregated to the industry (ISIC Rev. 4) level using the OECD's BTDIxE (Bilateral Trade Database by Industry and End-use) correspondence tables. These tables not only enable the crosswalk to ISIC Rev. 4 data that can be matched to the firm's industries, but also provide a first breakdown of products between intermediate and final goods.

Additional data is then used to decompose the intermediate goods subset of Chinese exports by importing industry. This is achieved through a two-fold estimation of sub-industry ratios following a proportionality assumption. Firstly, using the OECD's ICIO (Inter-Country Input Output) tables, I am able to identify, for a given year of Chinese industry i's exports, the proportion of intermediates going to each 2-digit ISIC industry. ICIO tables are constructed through the combination of national and international sources, especially national accounts data and national input-output or supply-use tables. Key to all input-output data, they produce a final estimation of both direct (input from industry i to industry j) and indirect inputs (industry j's inputs, and so forth) through the Leontief inverse matrix $\mathbf{B}$, with aims to have a full picture of an economy's linkages.

To achieve a 4-digit importing industry estimation, I require an additional decomposition of detailed supply-use tables (SUTs, which are of similar structure to input-output tables) for each country: Argentina's 2004 SUT, Brazil's 2011 SUT, and Mexico's 2008 SUT. This means that

the shares used to decompose importing industry are time-invariant. This is a common limitation within the literature: Acemoglu et al. (2016) and Taniguchi (2019), both discussing input access effects on local labour markets in the US and Japan, respectively, make use of a single-year input-output table, implicitly assuming these structures are unlikely to shift in the short-term. Since I am performing a two-tier estimation, based firstly on the ICIO, I argue that my assumptions are less constraining. Building on their approach, I assign constant shares exclusively for the composition of detailed industries' inputs (4-digit), related to the ICIO's time-variant share of aggregated industry input (2-digit).

Formally, let x be the exporting country, j the exporting industry (4-digit), c the importing country, a the importing (aggregate, 2-digit) industry, and i the importing industry (4-digit). Observing country x's exports of industry j to country c, I first discern final demand from inputs using BACI and BTDIxE; then decompose total inputs using aggregate industry's a (2-digit) shares of ICIO's total input exports from country x to country c; and finally, using national SUTs, I decompose aggregate a into the 4-digit industry i using i's share within a's total use of supplying industry j in country c.

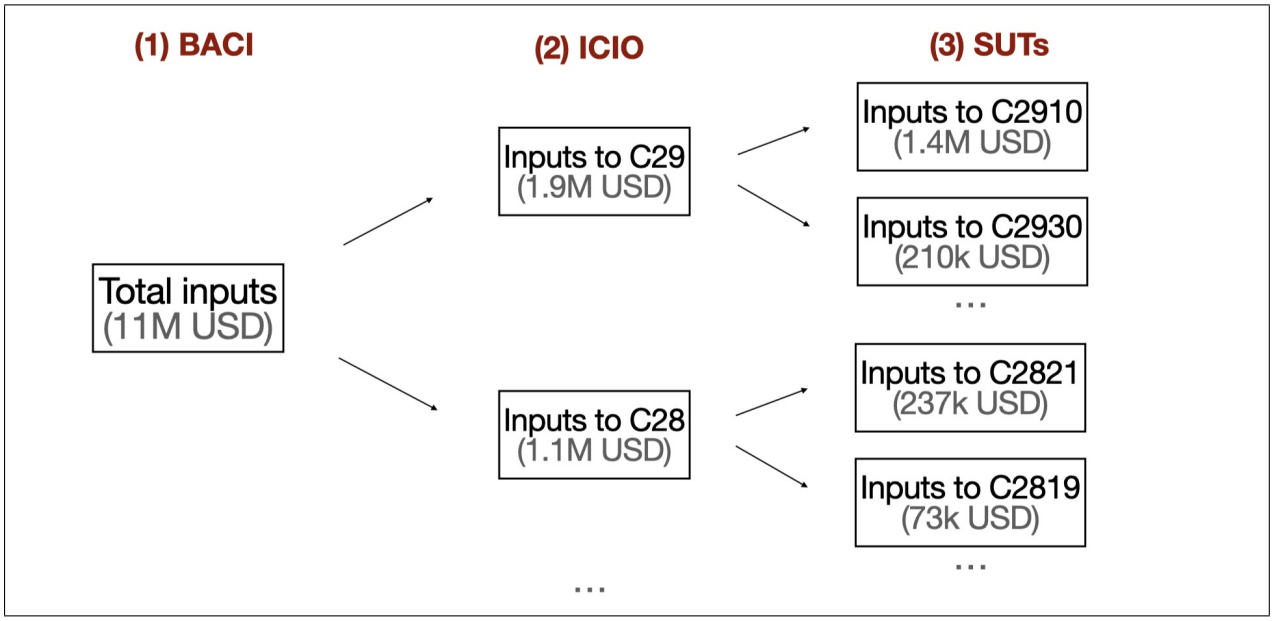
$$\begin{aligned} EXP_{x,c,j} &= EXP_{FD_{x,c,j}} + EXP_{INP_{x,c,j}} \\ &= EXP_{FD_{x,c,j}} + \sum_{n=1}^{a} EXP_{x,c,j,a} \\ &= EXP_{FD_{x,c,j}} + \sum_{z=1\in n}^{i\in n} EXP_{x,c,j,i} + ... + \sum_{z=1\in a}^{i\in a} EXP_{x,c,j,i} \end{aligned} \tag{2}$$

Thus, this final step of the estimation exclusively reflects an assumption of stable input mixes, which are more structural by nature. But the export flow from x, j to c, i is still variant according to the ICIO's allocation of input exports to aggregate a, where $i \in a$.

The result for this exercise can be exemplified in Figure 2. Where BACI and BTDIxE inform that, in 2002, Argentina imported around USD 11 million worth of inputs from Chinese machinery (ISIC's C28), ICIO provides a (time-varying) decomposition that informs that USD 1.9 million of these inputs were imported by Argentinian manufacture of motor vehicles (C29). Finally, through the (time-invariant) ratios identified through the Argentinian 2004 SUT, I am able to infer that USD 210 thousand, around 2% of all Chinese machinery intermediates imports, had as their in-

dustry destination the manufacture of parts and accessories for motor vehicles (C2930). From the perspective of the focal firm in this industry in Argentina, the import of inputs is the sum of all input exports to C2930 from any industry – manufacturing and non-manufacturing – in China.

Figure 2: From BACI to a country-industry bilateral dataset: an example with Argentinian imports from Chinese machinery (C28) in 2002



4.3 Descriptive statistics

The final dataset is composed of 1,899 innovating firms across Brazil, Argentina and Mexico that have produced 6,409 patents in the period of 1996-2008. As discussed, these three countries represent the vast majority of patenting activity in Latin America. However, they do not do so equally: the sample is mainly represented by Brazil, which concentrates 75% of patenting firms and 79% of observed patenting activity. Argentina and Mexico account for the remaining 13% and 11.9% of patenting firms and 7.9% and 13.6% of the patent counts, respectively.[3]

Table 1 presents the composition of the sample at the 2-digit industry level, ordered by the decreasing order of change in patenting between 1996-2001 and 2002-2008. Patenting activity is significantly concentrated around four industries (machinery and equipment, electrical equipment, other manufactured products and chemicals), which together account for 81.9% of all patents and 76.6% of all patenting firms. Machinery and equipment alone represents more than a third of each.

[3]Table A.2 shows the country breakdown of patents and patenting firms for Latin America.

Table 1: Final sample by industry

Industry (ISIC 2-d)	(1) Firms_n	(2) $\text{Firms}_{\%}$	(3) Pat_n	(4) $\text{Pat}_{\%}$	(5) ΔPat_{yavg}	(6) ΔIMP	(7) ΔEXP	(8) ΔIMPI
Chemicals	233	12.3%	832	13%	47.4	4.0	-0.4	2.2
ICT and electronics	175	9.2%	369	5.8%	23.3	22.6	0.4	4.9
Machinery and equipment	653	34.4%	2471	38.6%	19.3	9.9	-0.5	7.2
Fabricated metal products	78	4.1%	223	3.5%	8.0	14.6	-0.9	7.3
Motor vehicles	135	7.1%	413	6.4%	2.4	2.7	-1.7	12.8
Printing and reproduction	7	0.4%	9	0.1%	0.7	12.8	3.1	5.5
Food products	1	0.1%	1	0%	-0.2	1.8	-0.1	0.7
Rubber and plastics	5	0.3%	7	0.1%	-0.2	10.0	0.7	7.0
Other non-metallic mineral products	45	2.4%	146	2.3%	-1.4	15.4	-0.2	3.2
Other manufactured products	358	18.9%	882	13.8%	-2.1	13.6	0.2	8.6
Electrical equipment	209	11%	1056	16.5%	-4.5	21.4	-2.7	9.8
Total manufacturing	1899	100%	6409	100%	92.5	12.1	-0.8	7.3

Note: The table is in descending order of the variation in average yearly patents (column 5). This variation corresponds to the difference between the average yearly patents in 1996-2001 and 2002-2008. The last three deltas correspond to the change between 2002 and 2008 in China's partner share of Argentina's, Brazil's and Mexico's combined imports, exports and input imports. Industries (ISIC 2-digit) that did not report any patenting between 1996 and 2008 are not reported.

Specified in the next section, the change in China's partner share in Latin America's imports, exports and input imports is represented in the last three columns. These changes measure import penetration, market access and input access and vary substantially across industries. While the largest increase in import penetration from China was in ICT and electronics, an industry that experienced a high increase in innovation, its next two largest values were for top-four (in patenting activity) industries that produced fewer patents after 2002. Notably, for nearly all manufacturing, China's partner share decreased between 2002 and 2008, highlighting again that Latin America's experience with the China shock had no significant demand-driven market access effect for Latin American manufactured goods.

In the annex, table A.1 shows the distribution of variables at the firm level. Its main takeaway is that patents are highly concentrated across firms, with the 50th and 75th percentiles corresponding to 1 and 2 patents between 1996 and 2008, respectively, whereas the mean patenting is 3.4. This relates to the innovator premia, which, besides the well-known exporter premia, suggests innovating

firms are represented by a smaller set of bigger and more productive firms (Aghion et al., 2024). From this observation, a key fact is derived: most of the variation observed in the dataset reports firms that entered (in this context, did not patent in 1996-2001 and patented in 2002-2008) or exited (the opposite) the pool of innovators. The identification strategy to define this variation is described in the next section.

5 Empirical Strategy

The empirical analysis is based on a first-difference estimation with country-industry (aggregate) fixed effects, similar to the framework proposed by Autor et al. (2020).

$$\Delta Y_{i,c,j} = \beta_0 + \beta_1 \Delta IMP_{c,j}^{CHN} + \beta_2 \Delta EXP_{c,j}^{CHN} + \beta_3 IMPI_{c,j}^{CHN} + FE_{ca} + \varepsilon_{f,c,j} \tag{3}$$

The outcome variable $\Delta Y_{i,c,j}$ is the DHS growth rate of the number of innovative patents. It symmetrically defines the relative change of average yearly patent output (bounded by -2 and 2), while also being defined for firms that did not innovate in the pre-period (having the value of 2).[4]

$$\Delta Y_{f,c,j} = 2 \cdot \frac{\bar{p}_{f,c,j,t} - \bar{p}_{f,c,j,t-1}}{\bar{p}_{f,c,j,t} + \bar{p}_{f,c,j,t-1}} \tag{4}$$

where $\bar{p}_{f,c,j,t}$ defines the average yearly flows of patents for firm f in country c and industry i in the period of 2002-2008, and $\bar{p}_{f,c,j,t-1}$ analogously refers to the pre-period of 1996-2001.

The variables of interest are β_1, β_2 and β_3, which reveal the direction (positive or negative) and magnitude of the correlation between innovation and Chinese import competition, market access and input access, respectively. Country-industry fixed effects (FE_{ca})[5] purge any common time invariant shocks that could have happened in industries, such as managerial techniques or a particular high propensity to patent (e.g. the pharmaceutical industry). The first differencing also provides control for unobserved distinctions, such as firm-specific fixed effects.

[4] Haltiwanger et al. (2013) describe in their annex the full motivation and interpretation consequences of this method of estimating a growth rate

[5] Similar to the approach of Autor et al. (2020), the industries controlled for in the fixed effects are more aggregate than ISIC's 34 unique 2-digit manufacturing industries. Rather, I categorize 8 manufacturing aggregate industries of similar characteristics, following the OECD's ICIO. The full list is provided in table A.3 Although some studies prefer to use ISIC 2-digit fixed effects, results were more volatile and less significant, likely indicating the limitations of a smaller sample of Latin American patents and firms.

Following Ahn et al. (2018), I proxy for import penetration, market access and input access by measuring the variation of China's weight as partner for a given country-industry's imports, exports and input imports.

$$\Delta flow_{c,j}^{China} = \frac{flow_{c,j,2008}^{China}}{flow_{c,j,2008}^{World}} - \frac{flow_{c,j,2002}^{China}}{flow_{c,j,2002}^{World}} \tag{5}$$

where $flow$ corresponds to each of the three trade flows: $IMP_{c,j}^{CHN}$, $EXP_{c,j}^{CHN}$ and $IMPI_{c,j}^{CHN}$.

There is a well-known concern of endogeneity in this setting, related to potentially Brazil-, Argentina- or Mexico-driven components driving these flows. For example, the rise of China in Brazil's imports could be linked to a Brazilian industry-specific demand shock. An instrumental variable (IV) has to address this issue to isolate, i.e. in the case of import competition, the supply-driven shock of the China shock. Following Autor et al. (2020) and several papers that have implemented this strategy in their approach (Aghion et al., 2024; Ahn et al., 2018; Yamashita and Yamauchi, 2020), the instrument takes the following form:

$$\Delta flow_{Other,j}^{China} = \frac{flow_{Other,j,2008}^{China}}{flow_{Other,j,2008}^{World}} - \frac{flow_{Other,j,2002}^{China}}{flow_{Other,j,2002}^{World}} \tag{6}$$

where $flow_{Other,j}^{China}$ denotes the rise of China as a trade partner to similar countries to country c. This basket of countries in $Other$ includes large Latin American economies that exclude country c, that is, for Argentina, the basket is Brazil, Chile, Colombia, Mexico, Panama, Peru, and Uruguay. Because there are additional data coverage limitations for input-output data, $IMPI_{Other,j}^{China}$ draws from a rest of the world aggregate.

6 Results

6.1 Baseline estimates

Baseline estimates for the impacts of the China shock on patent production in Latin America are shown in Table 2, which presents an individual model for each trade flow and the full model in Columns (4) and (8). The OLS baseline estimation suggests that both import penetration and market access have had negative consequences to Latin American innovation. Findings on the latter suggest that the induced competition effect may have dominated the exports channel

in the case of Latin America, similar to what is found by Ghizzi (2021) for Brazil. For a 1p.p. change in China's share of Latin American imports, the annual average patenting is expected to decrease by 1.4%.[6] In the full model, these negative import penetration effects become stronger, but significantly counterbalanced by a positive effect of input access on innovation. Results suggest market access to China drove innovation downwards, although as shown, China never became a large market for Latin American manufacturing exports.

As discussed, the baseline OLS estimation suffers from endogeneity issues. For example, time-varying technology shocks in an industry could be correlated with the rise in Chinese imports. A Chinese supply-driven import penetration shock and demand-driven export market shock is captured with the rise of Chinese exports and imports to other, similar Latin American economies. For example, the rise of China as a major exporter to Peru is highly correlated with imports of Chinese goods from Brazil, but it is not associated with technology shocks in Brazilian industries. Results suggest this is particularly true for access to inputs, where neutralizing the demand-driven shock significantly increases the coefficient's magnitude and significance.

Although both the OLS and 2SLS models point to a negative effect of Chinese import competition and market access on patent production, less significance is observed in the 2SLS model (first-stage regression is presented in table A.4). The full 2SLS model (column 8) reveals a strong and significant effect of access to Chinese inputs on patent production, suggesting that trade liberalization with China and the penetration of Chinese inputs solidly enabled Latin American firms to learn and optimize costs. A 1p.p. increase in the import penetration of Chinese intermediates would lead to a 18% increase of average annual patenting in Latin American firms.

6.2 Industry heterogeneity

There is widespread evidence of the heterogeneous effects of import competition and market access on firms based on their size, productivity or sales (Aghion et al., 2024; Ahn et al., 2018; Autor et al., 2020). Regardless of the sense of the relationship between these shocks and firm performance, larger, more productive firms are in general better off than their smaller counterparts. I test in the dataset for the heterogeneous effects of technology intensity of these firm's sectors. This is following

[6] Bounded between -2 and 2, the coefficient approaches the regular percentage value for a growth rate around zero. As it approaches its bounds, a DHS growth rate no longer can be read as a regular growth rate. For reference, a -50% decrease relates to a -.067 DHS decrease, and a +1 DHS increase translates into a three-fold increase.

Table 2: Impact of Chinese import penetration, market access and input access on Latin American innovation

	OLS				2SLS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔIMP	-0.014^{**}			-0.018^{***}	-0.013^{*}			-0.010^{**}
	(0.006)			(0.005)	(0.007)			(0.004)
ΔIMPI		0.040		0.056^{**}		0.151^{***}		0.163^{***}
		(0.028)		(0.026)		(0.040)		(0.031)
ΔEXP			-0.016^{***}	-0.015^{***}			-0.008	-0.020^{**}
			(0.005)	(0.004)			(0.010)	(0.009)
Num. obs.	1899	1899	1899	1899	1899	1899	1899	1899
Mean outcome variable	0.275	0.275	0.275	0.275	0.275	0.275	0.275	0.275
R^2	0.057	0.055	0.056	0.065	0.056	0.064	0.053	0.070
F-stat. (first-stage)					65.0	18.3	11.4	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

a categorization from Eurostat, where manufacturing sectors at the 3-digit level are defined as either high-, medium-high-, medium-low-, and low-technology. I consider a category of high technology, comprising the former two definitions, and low technology, which correspond to the latter two.

Table 3 splits Latin American firms into these two groups. Firstly, conducting the OLS analysis, I lose significance for all trade flows for firms in low-technology sectors, and results are being driven by high-technology firms. This narrative shifts once the IV analysis is conducted. The negative effect of market access on innovation is largely driven by less technology-intensive firms. Most clearly, what emerges from results are the positive and significant coefficients for the impact of input access on innovation on firms in high-technology industries. These firms seem to have reaped the most benefits from efficiency gains and learning from the inputs imported from China.

Complementary to this narrative, table A.5 presents results where the rise of China's weight as a trade partner for Latin American input imports is split by high- and low-technology exporting industries. The table sheds light on the discussion of which inputs might matter most in the Latin American context. Observing the full 2SLS model (column 4), it is clear that what drove most innovation in Latin American was the access to high-technology inputs, emphasizing the learning channel rather than the cheaper inputs channel. An interesting observation is the strong negative effects of low-technology inputs imports penetration in the OLS estimates, which completely loses significance in the 2SLS approach. This highlights the potential endogeneity issues previously

Table 3: The China shock and industry heterogeneity

	OLS		2SLS	
	(1)	(2)	(3)	(4)
	Low tech	High tech	Low tech	High tech
ΔIMP	−0.027	−0.017*	0.007	−0.007
	(0.016)	(0.010)	(0.027)	(0.006)
ΔIMPI	−0.006	0.063**	0.109	0.169***
	(0.322)	(0.028)	(0.199)	(0.031)
ΔEXP	−0.030	−0.015***	−0.330*	−0.021**
	(0.102)	(0.004)	(0.188)	(0.010)
Num. obs.	362	1537	362	1537
Mean outcome variable	0.072	0.323	0.072	0.323
R^2	0.061	0.043	0.065	0.049

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. First stage for instrumental variables is based on the previous estimation, for individual flows and including the full sample.

discussed, where this influx of low-technology inputs might be correlated to increased demand from less innovative firms.

6.3 Robustness checks and alternative specifications

Up to this point, all firms have been treated equally in the estimations. However, one potential setback of this study is that results are mainly driven by firms that have had participated only minimally in innovation. This can be particularly true in the Latin American context, as patenting is much scarcer than in other regions. With DHS growth rates, the largest variation in outcome are firms that have either started or stopped innovation between 1996-2001 and 2002-2008. Out of all 1,899 firms in the dataset, 88% have a 2 or -2 DHS growth, and 61% have only patented once between 1996 and 2008. Figure A.1 further illustrates this observation.

As a robustness check, I follow Autor et al. (2020) and report results weighted by patent count in 4. The results remain consistent with the baseline model, with the notable increase in magnitude of input access on innovation. The full IV patent-weighed model suggests that 1p.p. increase in access to Chinese inputs drives a 25% increase in granted patents.

Additionally, I check for the lawyer effect to verify if this increased activity in patenting led to lower quality patents, applied for and granted mainly to protect already existing intellectual

Table 4: Patent-weighted estimation

	OLS				2SLS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔIMP	-0.014^{**}			-0.018^{***}	-0.013^{*}			-0.008^{*}
	(0.006)			(0.005)	(0.007)			(0.004)
ΔIMPI		0.043		0.064^{***}		0.178^{***}		0.208^{***}
		(0.028)		(0.024)		(0.041)		(0.027)
ΔEXP			-0.013^{***}	-0.011^{***}			-0.005	-0.018^{*}
			(0.004)	(0.004)			(0.010)	(0.009)
Num. obs.	1899	1899	1899	1899	1899	1899	1899	1899
Patent-weighted average of outcome variable	0.286	0.286	0.286	0.286	0.286	0.286	0.286	0.286
R^2	0.052	0.049	0.049	0.061	0.047	0.061	0.047	0.065
F-stat. (1st stage)					66.4	19.1	10.8	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. All models are weighted by the log stock of patents of a firm between 1996 and 2008.

property rights. Following Bloom et al. (2016), I measure citation per patent as an outcome to control for any variation in patent quality, specifically its DHS variation as with patents. Results are reported table A.6 and add to the evidence that there has been no lawyer effect dominating the reaction of patenting firms to the China shock. Conversely, I report additional evidence that input access has been beneficial to firm performance in Latin America, suggesting that not only has it enabled them to innovate more, but has also enabled higher quality innovations.

Finally, I proceed with two alternative measurements to address potential threats to my identification strategy, related mostly to potential measurement errors. Firstly, I calibrate the model with a different input-output database, namely the WIOD (World Input-Output Database). Similar to the ICIO in its analytical potential, this database aims to provide a comprehensive and harmonized input-output database, covering 40 countries, 35 sectors and the years 1995 through 2011. It has been specifically designed to observe developments over time, as it is benchmarked to time-series of several national accounts components (Timmer et al., 2015). The disaggregation to ISIC 4-digit bilateral trade estimations was performed using the same supply and use tables, following the same procedure as the baseline model.

Since WIOD does not cover Argentina, in table A.7 I compare the input access instrument constructed through the ICIO and its WIOD counterpart for Mexico and Brazil. This loss of information (13% of observations corresponding to Argentinian patenting firms) does not significantly

affect the baseline results for the effect of input access on innovation. Both instruments are robust and the WIOD-constructed instrument, while at still large but smaller magnitude, reiterates the main findings that access to Chinese intermediates has significantly led to more innovation in Latin America.

These results are also robust to a re-definition of the selected economies to construct the instrument. In this exercise, I still follow Autor et al. (2020)'s rationale of instrumenting a supply-driven (demand-driven) shock from the rise of China as an important source of manufactured imports (exports) in Latin America by measuring its rise through other economies than country c. However, instead of using other Latin American economies for imports and exports and rest of the world for inputs, I use a basket of six emerging economies outside of the region: India, Indonesia, South Africa, Thailand, Turkey and Vietnam. The rationale for this approach is to further isolate the supply- and demand-driven shocks from any regional shock that might have been similar across Latin America.

Results in table A.8 are robust to this alternative basket of countries. An important feature of this regression is that exports flows cannot be included, because neither this basket of countries or rest of the world can robustly predict variation in China's share of Latin American manufactured exports. This reintroduces an import limitation of this study: the demand aspect of the China shock was perceived in Latin America as a commodity boom, with a less significant rise in demand for manufactured goods relative to the world. This "commodities-for-manufactures" boom translates into few variation of China's share of Latin American manufacturing, an experience not necessarily shared across other large, even emerging economies.

7 Conclusion

This paper contributes to the growing literature on trade liberalization and firm innovation by analyzing the impact of the China shock on innovation in Latin America. By disentangling the effects of import competition, market access, and input access, this study provides a nuanced understanding of how firms in emerging economies respond to large-scale trade shocks. The empirical findings suggest that these three forms of trade exposure have competing consequences in the case of Latin America: while increased import competition and market access to China exerted negative pressures on patenting activity, greater access to Chinese inputs significantly enhanced firm-level

innovation.

Similar to current evidence in the literature on firm heterogeneity, I find that firms present in or importing inputs from industries of higher technological intensity are the main drivers of these results. They tend to suffer the negative effects of increased Chinese import penetration and market access less intensely. At the same time, they are the main drivers of the input access effect, benefiting from efficiency gains and knowledge spillovers that have not only led to increased patenting activity, but also increased quality of innovation. This effect underscores the importance of trade liberalization not merely as a driver of competition but also as a channel for technological advancement.

The main contribution of this paper is the robust evidence on the positive effect of input access to innovation in emerging economies. While Aghion et al. (2024) find less significance in results for this channel in the context of the French response to the China shock, there is theoretical reasoning to suggest that emerging economies might have had stronger benefits from access to Chinese inputs. Since results are driven by high-technology input imports, findings suggest the main mechanism at play is learning and technology transfer. As Latin America stands on average further from the technology frontier in most industries compared to developed economies, there is larger potential to be tapped from Chinese inputs. In this context, in contrast to Aghion et al. (2024), my results for input access are unequivocally positive.

Disentangling input access from import competition is also relevant to understand the ongoing academic discussion on whether Chinese import competition has been benefitial or detrimental to firm innovation. Beyond the competing Schumpeterian and escape competition effects, the contrasting findings in Bloom et al. (2016) compared to Autor et al. (2020) should be partially explained by the confounding effects of input access and import competition captured by a single measure of Chinese import penetration. Different input-output structures might reveal different weights of each effect in the overall import penetration. Where in one economy most imports can represent firms sourcing inputs from China, in another these imports can reflect mainly final products going to consumers and competing with a firm in the same sector. These differences can be large drivers of the direction and magnitude of coefficients found in studies that do not discern these forms of trade exposure.

Overall, this study provides new insights into the role of trade exposure in shaping innovation dynamics in emerging markets. While trade liberalization can create opportunities for technological

advancement, it also presents challenges that require proactive policy responses. The findings highlight the importance of balancing competitive pressures with strategic support mechanisms to ensure that firms in developing economies can leverage globalization for sustainable innovation and growth. Observing the shock episode itself, the China shock in Latin America has to be understood as a commodities-for-manufactures trade boom. As such, its negative effects of market access were largely muted, as China's rise in international trade never resulted in a disproportionate rise in demand for Latin American manufactured goods. I find that its main impact on Latin American patenting was an innovation-inducing influx of intermediates.

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A Appendix

Figure A.1: Distribution of firm total patent stock and DHS delta in the sample

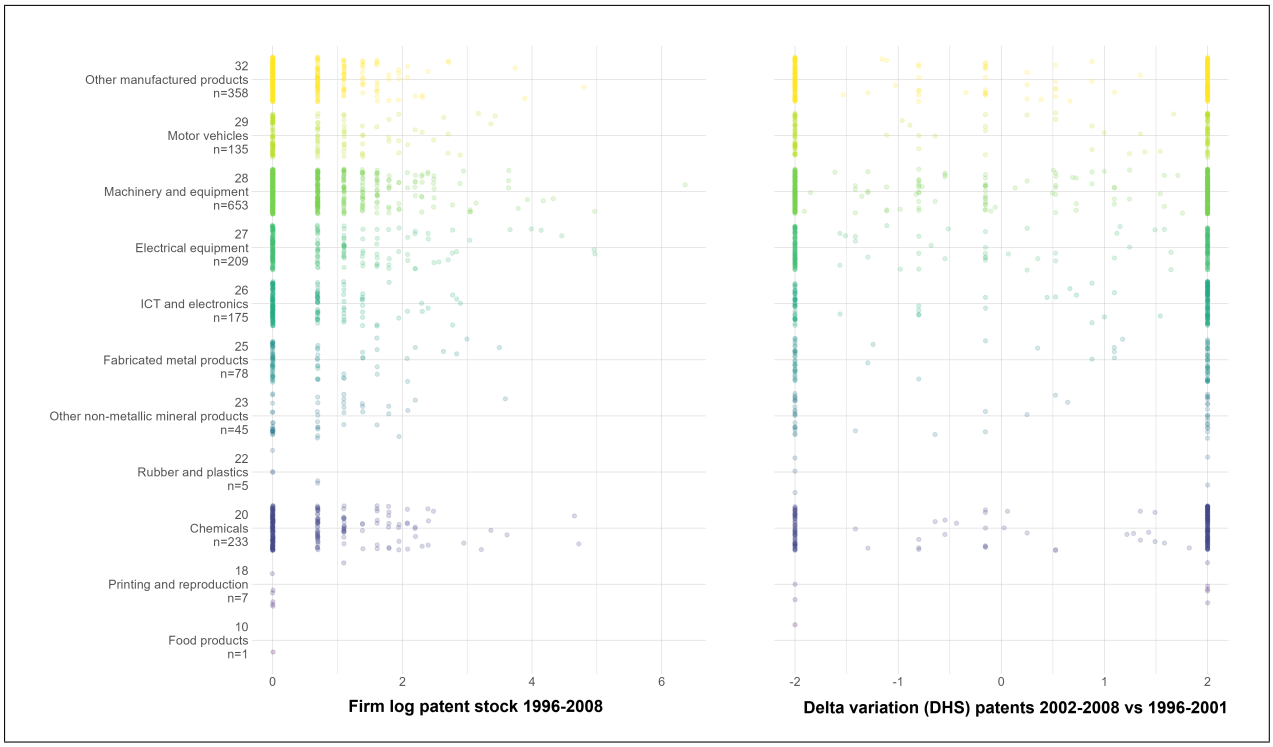

Table A.1: Summary of sample distribution at the firm level

Variable	(1) Mean	(2) SD	(3) 10th	(4) 25th	(5) 50th	(6) 75th	(7) 90th
Pat	3.4	16.0	1.0	1.0	1.0	2.0	5.0
ΔPat	0.3	1.9	-2.0	-2.0	2.0	2.0	2.0
Cit	25.0	176.7	0.0	0.0	0.0	10.0	38.0
ΔCit	0.2	1.2	-2.0	0.0	0.0	1.3	2.0
ΔIMP	13.1	11.0	2.3	4.5	8.9	16.4	29.6
ΔEXP	-1.3	6.9	-1.8	-1.1	0.0	0.2	0.9
ΔIMPI	8.1	3.2	4.4	7.1	7.8	9.2	11.2

Note: Deltas for outcome variables patents (column 2) and citations (column 4) follow the specification of the DHS growth rate used in the empirical strategy. The DHS growth rate is symmetric and allows for defining variation of entry (i.e., started patenting, defined at 2) and exit (i.e., stopped patenting, defined at -2).

Table A.2: Distribution of patenting firms and patents by Latin American country, 1996-2008

	PATSTAT				Final sample			
Country	Firms_n	$\text{Firms}_\%$	Pat_n	$\text{Pat}_\%$	Firms_n	$\text{Firms}_\%$	Pat_n	$\text{Pat}_\%$
Argentina	387	12.2%	823	8.7%	247	13%	504	7.9%
Brazil	1872	59.1%	6343	66.8%	1426	75.1%	5036	78.5%
Mexico	366	11.6%	1080	11.4%	226	11.9%	869	13.6%
LATAM3	2625	82.9%	8246	86.9%	1899	100%	6409	100%
Rest of LATAM	541	17.1%	1243	13.1%	-	-	-	-
LATAM	3166	100%	9489	100%	-	-	-	-

Source: PATSTAT
Notes: The difference between PATSTAT and final sample in values for Argentina, Brazil and Mexico arise from multiple steps in data processing. Firstly, the number of firms from PATSTAT is overestimated due to different names and typos in the dataset, which are treated through name matching exercises described in the data section. Secondly, the industry cannot be accurately estimated for all firms, due to eventually incomplete data in PATSTAT. Finally, a small portion of these firms are innovating outside manufacturing and are not part of the object of this study.

Table A.3: The eight industry aggregates under manufacturing

Industry aggregate	ISIC 2-digit industries
Food and tobacco	10 to 12
Textiles and apparel	13 to 15
Wood, paper and printing	16 to 18
Chemicals and non-metallic mineral products	19 to 23
Basic and fabricated metals	24 to 25
Machinery, equipment and electronics	26 to 28
Transport equipment	29 to 30
Other manufacturing	31 to 33

Note: Industries aggregates derived from the breakdown provided by the ICIO-derived (Inter Country Input-Output Tables) Trade in Value Added database (oe.cd/tiva).

Table A.4: Baseline estimation, first-stage regression

	$\Delta\text{IMP}_{LATAMother}$	ΔIMP_{ROW}	$\Delta\text{EXP}_{LATAMother}$
ΔIMP	1.287***		
	(0.079)		
ΔIMPI		1.267***	
		(0.345)	
ΔEXP			1.234***
			(0.385)
Num. obs.	1899	1899	1899
R^2	0.917	0.757	0.660
F-stat. (first-stage)	65.0	18.3	11.4

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

Table A.5: The China shock and input access heterogeneity

	OLS		2SLS	
	(1)	(2)	(3)	(4)
$\Delta\text{IMPI}_{High-tech}$	0.037^{**}	0.045^{***}	0.087^{***}	0.094^{***}
	(0.017)	(0.013)	(0.028)	(0.027)
$\Delta\text{IMPI}_{Low-tech}$	-0.084^{**}	-0.104^{***}	-0.032	-0.030
	(0.039)	(0.019)	(0.086)	(0.074)
ΔIMP		-0.011^{***}		-0.011^{**}
		(0.004)		(0.005)
ΔEXP		-0.029^{***}		-0.019^{*}
		(0.004)		(0.010)
Mean outcome variable	0.275	0.275	0.275	0.275
Num. obs.	1899	1899	1899	1899
R^2	0.041	0.055	0.042	0.048
F-stat. (first-stage)			16.0 (HT) 25.6 (LT)	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

Table A.6: The China shock and citations per patent

	OLS				2SLS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔIMP	-0.001			-0.001	0.000			0.001
	(0.003)			(0.003)	(0.003)			(0.003)
ΔIMPI		0.006		0.007		0.049^{**}		0.052^{**}
		(0.013)		(0.012)		(0.020)		(0.022)
ΔEXP			-0.002	-0.002			0.002	-0.002
			(0.003)	(0.003)			(0.003)	(0.003)
Num. obs.	1844	1844	1844	1844	1844	1844	1844	1844
Mean outcome variable	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.228
R^2	0.019	0.019	0.019	0.019	0.020	0.022	0.019	0.022
F-stat. (1st stage)					61.6	17.3	10.8	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

Table A.7: The input access effect, calibrated with the ICIO and WIOD

	First-stage		Second-stage	
	(1)	(2)	(3)	(4)
	ICIO	WIOD	ICIO	WIOD
ΔIV-IMPI	1.236***	1.456***		
	(0.450)	(0.142)		
ΔIMPI			0.155***	0.071**
			(0.048)	(0.034)
Num. obs.	1652	1652	1652	1652
Mean outcome variable	0.373	0.373	0.373	0.373
R^2	0.656	0.851	0.023	0.016
F-stat. (first-stage)	11.8	35.3		

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

Table A.8: Alternative basket of countries: six non-Latin American emerging economies

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	−0.014*		−0.009*	−0.017*		−0.022***
	(0.008)		(0.005)	(0.009)		(0.008)
ΔIMPI		0.165***	0.154***		0.080**	0.105***
		(0.041)	(0.036)		(0.039)	(0.032)
Num. obs.	1899	1899	1899	1899	1899	1899
Mean outcome variable	0.275	0.275	0.275	0.275	0.275	0.275
R^2	0.033	0.043	0.045	0.033	0.033	0.040
F-stat. (first-stage)				13.6	13.8	

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs are clustered on 4-digit country-industries. All columns contain country-industry fixed effects, where industries are split into 8 aggregates: food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. The alternative basket of countries for the instrumental variable is composed of India, Indonesia, South Africa, Thailand, Turkey and Vietnam.